

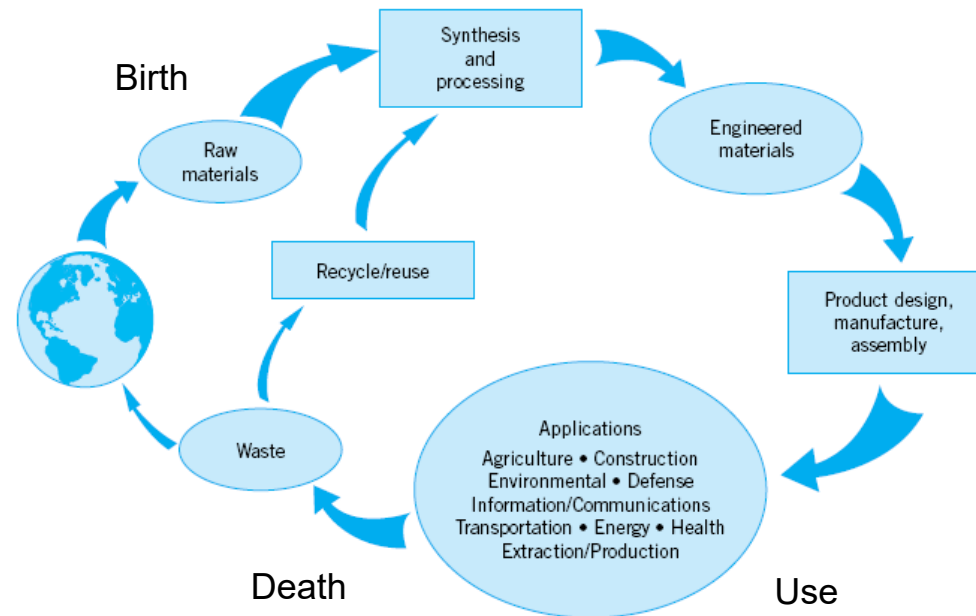
ENGEN370 Environmental Impact Assessment (Six)

Environmental Management – ISO14001

- ISO , the International Organisation for Standardisation, is a world wide federation of national standards bodies
- The ISO14000 family is a group of standards/tools created to help companies and organisations manage their environmental responsibilities
- These include auditing, communications, labelling, management and life cycle analysis
- ISO14001 sets out criteria for an environmental management system
- Companies/organisations can be certified under ISO14001
- Why do we need an environmental management system?

Materials Life Cycle

- 15 billion tonnes of raw material extracted from earth each year



Maturity



Construction Industry

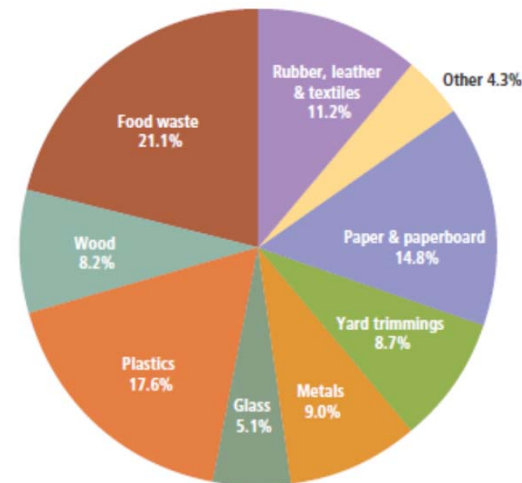
- European Union
 - 859 million tonnes of waste, 38% of total waste
- China
 - 45% of total energy consumption
 - 25% of total solid waste
- NZ
 - 50% of total solid waste



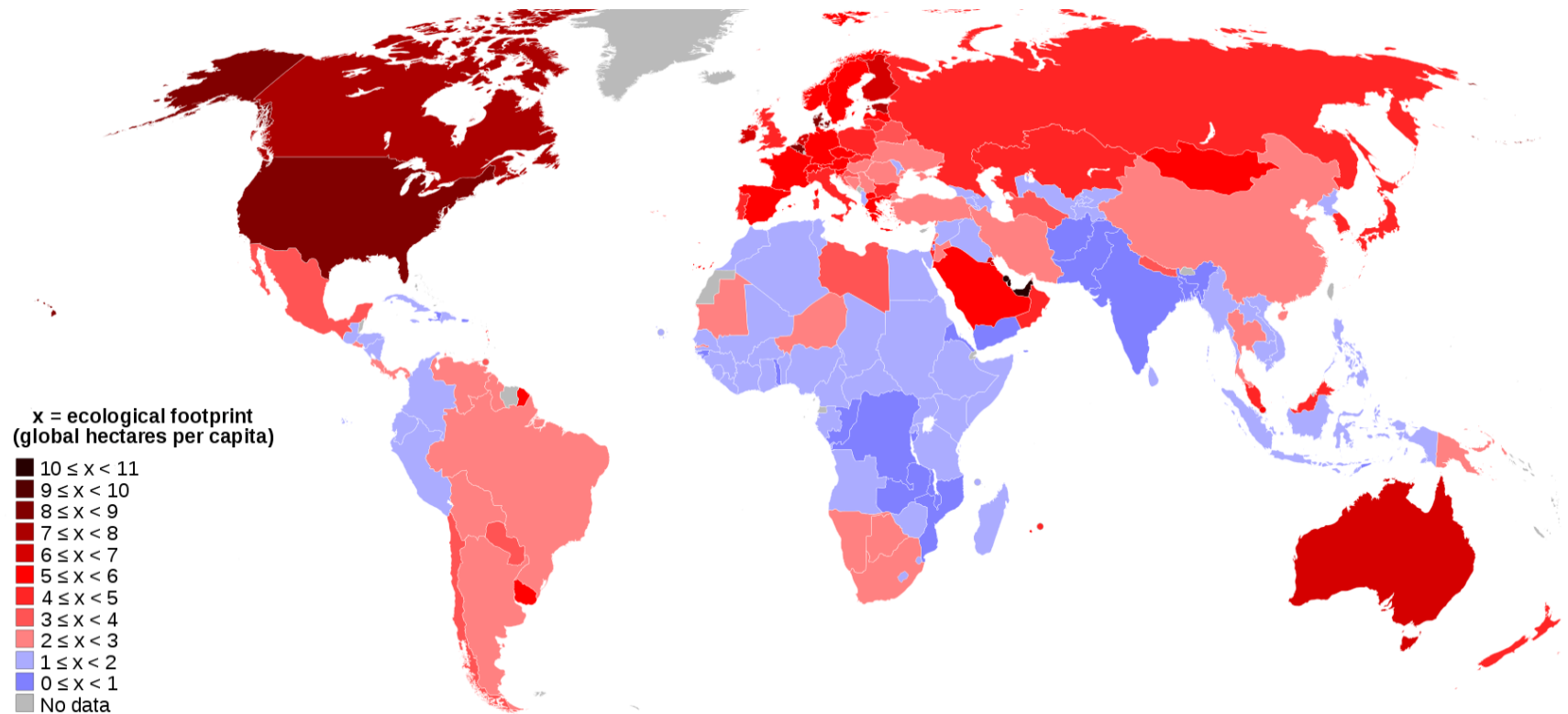
Municipal Solid Waste

- US – 250 million tonnes per year
 - 65 million tonnes recycled
 - 21 million tonnes composted
 - 29 million tonnes combusted
- NZ – 2.5 million tonnes per year
 - 75% could be potentially recycled, recovered, reused

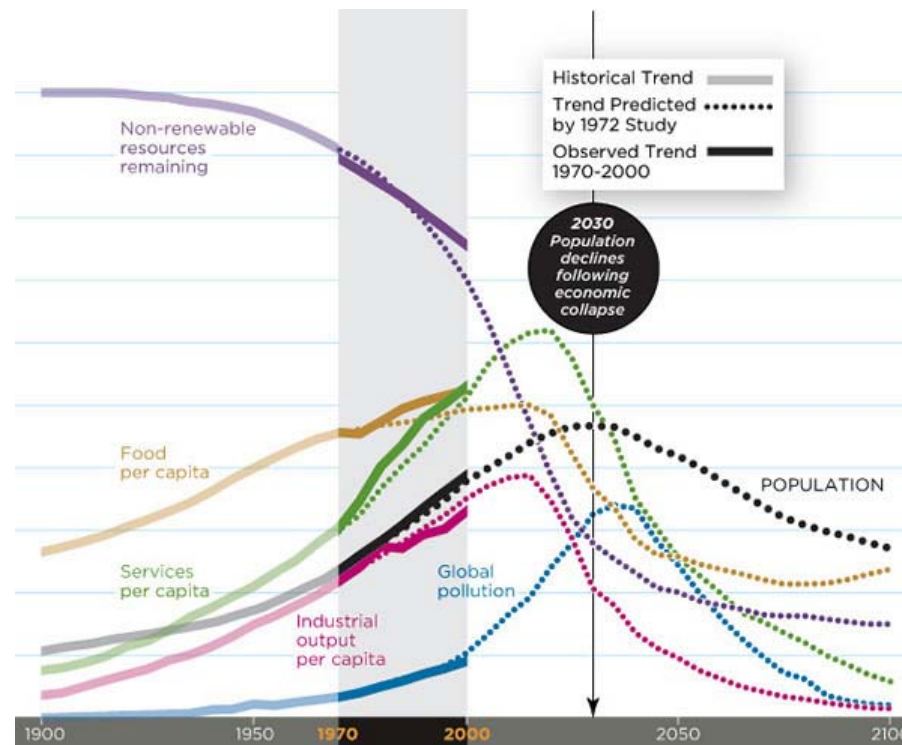
Figure 7. Total MSW Discards (by material), 2012
164 Million Tons (after recycling and composting)



Ecological Footprint – 1.7x Earth



Economic Collapse?



1972 study by Massachusetts Institute of Technology
"Limits of Growth", compared with current findings by
Graham Turner – Australian Physicist

Environmental Disasters

- Minamata disease (mercury poisoning from wastewater), Japan
- Itai-itai disease (cadmium poisoning from mining wastes), Japan
- Love Canal, Niagara Falls, (toxic waste dumped in a landfill), New York
- Seveso disaster (aerial release of 6 tonnes of toxic chemicals) , Italy
- Bhopal disaster (gas leak containing methyl isocyanate from a pesticide plant), India
- Phillips disaster (explosion/fire at a chemical plant) , Texas
- Tailings dam collapse (1000km of river affected), Ok Tedi, PNG
- Centralia coal seam fire (started in 1962, could still be burning >200 years from now), Illinois
- Deepwater Horizon (drilling rig explosion), Gulf of Mexico

Sustainable Development

- “To harmonise economic, social and environmental policies to ensure socially responsible economic development while protecting the resource base and environment for future generations” – Agenda 21, Earth Summit, 1992
- Contributors to the concept:
 - United Nations Conference on Human Development (1972)
 - “World Conservation Strategy” International Union for the conservation of nature and natural resources (1980) – proposed sustainable development
 - “Our Common Future” Brundtland Report, World Commission on environmental development (1987) – defined sustainable development

Sustainability

- To preserve what we have
- To reduce waste
- To reduce environmental impact
- To increase efficiency of use
- To maintain economic development
- To maintain quality of life

Daly Rules

Chief Economist World Bank, Herman Daly, 1971, "The entropy law and the economic process"

- Renewable resources such as fish, soil and groundwater must be used no faster than the rate at which they regenerate
- Non-renewable resources such as minerals and fossil fuels must be used no faster than renewable substitutes for them can be put in place
- Pollution and wastes must be emitted no faster than natural systems can absorb them, recycle them, or render them harmless

ISO 14000

- Why do we need an environmental management system?

Hopefully answered this question